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ANYTHING IS POSSIBLE

ABSTRACT. This paper criticises necessitarianism, the thesis that there is at least one necessary truth; and defends possibilism, the thesis that all propositions are contingent, or that anything is possible. The second section maintains that no good conventionalist account of necessity is available, while the third section criticises model theoretic necessitarianism. The fourth section sketches some recent technical work on nonclassical logic, with the aim of weakening necessitarian intuitions and strengthening possibilist intuitions. The fifth section considers several *a prioristic* attempts at demonstrating that there is at least one necessary proposition and finds them inadequate. The final section emphasises the epistemic aspect of possibilism.

1. POSSIBILISM

I begin with the thesis of *possibilism*, by which I mean the group of theses that all propositions are possible, or possibly true, that all propositions are contingent, that no proposition is necessary. The denial of the latter is the thesis of *necessitarianism*, the thesis that at least one proposition is necessarily true. It will be maintained in this paper that necessitarianism is false and that possibilism is true.

One might postulate a link between possibilism and the idea of *epistemic monism*, that there is only one basic epistemic method for investigating the world. It is invariably difficult to say what is monistic about monisms, since there can be disagreement about what is a 'basic' category. Again, within good scientific method (of which there are many different accounts) one can discern distinct roles for theory and sensory experience; though it is common currency these days to acknowledge a unity within scientific method between adjusting theory to sensory information, and adjusting the interpretation of the senses to well-constructed theory. But philosophers have usually shown a preference for monisms over dualisms, a preference which I share and which I think should be recognized in the theory of method, epistemology, as much as anywhere else. When Popper, for example, strayed from his general fallibilism, it was because he supposed a fundamentally different method for establishing logical truth; and Lakatos' criticism of Popper's conventionalism about both logic and mathematics was in the name of a properly general fallibilism.[1]

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The first question to ask is about the sources of necessitarianism, and this will be pursued in the following sections. I postulate two sources of necessitarianism, conventionalist necessitarianism and model theoretic necessitarianism, and in the next two sections these are discussed and rejected. In later sections, I endeavour to persuade the reader to abandon necessitarianism altogether. In the final section, I return to the question of the link between possibilism and epistemology, and maintain that only possibilism is the natural epistemological direction to take. One final point: intuitions about possibility seem to be rather easier to come by than intuitions about necessity. It seems to me that we have a strong sense that *some falsehoods are yet possible*, and that is my starting point.

2. CONVENTIONALIST NECESSITARIANISM

Popper and Lakatos objected to the *strategy* of conventionalism, by which they meant the method of preserving a thesis come what may from empirical refutation or criticism. Thus, *qua* strategy, conventionalism about any proposition violates correct fallibilist practice. But conventionalism, particularly conventionalism about logical truth, lends itself to a group of deeper, semantical theses as well. The distinctive appeal to *convention* is an appeal which following Popper,[2] we can characterise as *controllability in principle by decision*. We need to understand the effect this has on necessitarianism. In maintaining that necessary propositions form a nonnull class, conventionalist necessitarianisms have held that a convention can sometimes suffice to make a proposition true, and sometimes not. This evidently raises the question of how such a thing could be and that is discussed presently. Contrary to Wachbroit,[3] some version of conventionalist necessitarianism seems still to be believed by many philosophers, who were taught it as twentieth century orthodoxy. On the other hand, I would agree with Wachbroit that many philosophers do seem to think that Quine's 'Truth by Convention' is decisive against conventionalist necessitarianism, and that the latter is not obviously so.[4] Susan Haack argues that Quine's changing views on the status of the laws of logic are far from satisfactory;[5] and certainly Quine's strategy in 'Truth by Convention' of requiring conformity to the ordinary language meanings of the connectives is as conservative and conventionalist in

strategic effect as his later arguments in *Word and Object*.[6] Conventionalism needs re-killing.

It is important not to confuse conventionalism here with what Grünbaum called *trivial semantic conventionalism*. A simple distinction between sentences and propositions serves to make the point. Evidently the relationship between symbols and their meanings is in some sense controllable by a decision to associate a word with one semantic item rather than another. But this fact is entirely compatible with all propositions enjoying an equal epistemic status, since indeed it is a common feature of all sentences. Such semantically based conventionalism is thus trivial. It does not suffice to sustain a distinction between logical truths and fallible, empirical truths. Trivial semantic conventionalism seems to be behind the view of many philosophers that you can create logical truths simply by stipulations or resolutions that words have one meaning rather than another. But it should be apparent that it precisely does not serve to make a difference between logical and contingent truth. What would be needed for that would be a prior distinction between those kinds of meanings or propositions which, when fixed, *ipso facto* have a truth value; as opposed to those propositions whose truth value also varies with, or is determined by, the world. Grünbaum himself, as is well known, espoused metric conventionalism, which he explicitly denied was trivial. (See [4]). He argued that *even* given satisfactory semantics for metric propositions, a continuous space is *indeterminate* with respect to those metrical features, as demonstrated by the existence of alternative, incompatible metrical descriptions of it. Note the distinctive indeterminateness thesis. If a decision beyond a trivial semantic convention is needed to (or able to) determine truth, then the world must not determine the truth of the proposition in question.[7] We should be clear that it is not simply that there is something wrong with an indeterminateness thesis about a class of propositions, since presumably a thesis such as Grünbaum's is at least intelligible.[8] But it is *the coupling of an indeterminateness thesis with the further claim that such propositions can have their truth determined by decisions*. It is not easy to understand in what the *truth* of a proposition, with respect to which the world played no determining role, could consist. I take it that this objection applies particularly severely to conventionalisms about the propositions of logic and mathematics. After all, how could a decision (more generally, a convention) *conceivably* make something true if the world

plays no determinative role? One might feel inclined to wonder whether it was like printing several copies of the morning paper in order to make it true. Hilary Putnam puts this nicely: 'To put it bluntly, you can't make the Principle of Contradiction true by convention unless it's *already true*'.[9] Again, one might ask of any conventionalism including Grünbaum's, what could justify the use of the word 'true' of propositions with respect to which the underlying universe is indeterminate. If such are controllable by decision and escape control by an indeterminate world (while other propositions do not), then it seems odd that the word 'true' would be worth using of them at all. That way lies pragmatism.

Quine's changing views on logical truth mean that one must be careful in specifying just which parts one disagrees with. First, for all his apparent gradualism and fallibilism, Quine usually did give classical first order logic a unique role to play in theory, a role which I think has actually served to stifle inquiry into nonclassical logics in the recent past. This is, simply, conventionalist strategy. In fairness to Quine, his views on the conservation of classical logic were supported by the well-known arguments about radical translatability which I will not discuss here; though to the extent that they amount to a sophisticated version of the philosopher's 'I do not understand', more will be said in the next section. Second, while I do not deny that there can be an intelligible decision *never to give up* a certain class of propositions, that is a very different matter.[10] With this, my quarrel is rather different, namely that it is silly to make such decisions. Unless differences in the world under-determine differences in the truth value of the proposition in question, then to make a decision to believe a proposition in advance of ordinary epistemic investigation is to close one's eyes to the possibility of revision, as Quine the fallibilist insisted on other matters. Third, while Quine shied away from the notions of necessity, possibility and analyticity, he certainly thought early and late that conformity to classical logic and in particular to classical consistency was a rigid constraint on theory. That is *some* kind of impossibility thesis: it is one thing to hold that a proposition is inconsistent but something stronger to assert with Quine that this is invariably sufficient for its prohibition. On the other hand, I do not propose an account here of whatever it is that possibility consists in. The aim is, rather, to remain neutral on various of the going accounts,

for instance various realisms, and to argue for lifting bounds on the *extent* of possibility.

The earlier distinction between propositions which escape control by the world and propositions which do not, sat uncomfortably with the positivist-conventionalists, who often tried to deal with it with some version of the thesis that analyticities were not 'cognitively meaningful'. A typical conventionalist manoeuvre here is to deny that analyticities are in any straightforward sense true or false, that necessary truth is a kind of truth.[11] Popper's positivist links show here, for example, with his descriptions of them as 'truisms' and 'empty'; though his most favoured terminology is that also favoured by intuition, that 'All cats are cats' and its ilk are, simply, true. The problem of the nature of necessity arises just because some *truths* have seemed to have a special status.

I conclude that conventionalist necessitarianism is in serious trouble, both epistemically, and with its semantic underpinnings. However, mention of the matter of the literal truth of necessary truths brings me to my second classification of necessitarians. If necessary truth really is truth, then what sort of literal truth is it? One answer which many have been inclined to give, is that it is in virtue of being true in all members of some class of models that propositions are necessary. I call this model-theoretic necessitarianism, and in the next section it will be criticised and rejected.

3. MODEL THEORETIC NECESSITARIANISM

The view considered in this section is that necessity (and, let it be added, validity[12]) arises in virtue of playing a distinctive role in all models or semantical assignments. To avoid conventionalist problems about 'truisms', it is frequently claimed that the existence of models and truth preservation is a perfectly objective phenomenon, prior to mere syntax. Now one quick point to make is that this is frequently in an even worse position than conventionalism over its epistemology. At least conventionalism makes a semblance of squaring itself with its own epistemology, in that *declaring a word to have a meaning* is an event we seem to be able to get into epistemic contact with, even if the latter saddles us with the magical power of a decision to make something true when the world is supposedly unable to. But if neces-

sary truth and validity reside in timeless objective models (for instance the possible worlds of the modal realists), then we should want to know with which faculty we are able to get an inkling into their existence. More will be said on this in the next section. Here I want to put the criticism that model-theoretic necessitarians either take too simplistic a picture of what structures are available to them, or saddle themselves with an unarguable and uncriticisable metaphysics.

The fact is, that modern semantics has gone far beyond the simple models of classical first-order logic. Intuitionist models give up Excluded Middle, Paraconsistentist models give up the law of Non-contradiction, the various models of *Relevant Logics and their Rivals* [25] split apart vast numbers of logical theses which are equivalent in the context of classical logic, showing how to maintain one without the other. Some of these are considered in more detail in the next section. The semantics of classical first order logic is very much a special case of a much wider semantical framework, many of the details of which have only become apparent recently. I do not maintain the thesis that the *mere* existence of alternative models *suffices* to demonstrate that propositions refuted therein are not really necessary truths. I am making the weaker point that a model-theoretic necessitarian is in no position to say that various theses are logically true *solely* on the grounds that they hold in all models. Therefore, the model-theoretic necessitarian is in the position of having to argue that certain models have a *preferred* status over others. But it is not easy to see how this is to be done short of metaphysical dogmatism. The model theory *by itself* does not provide this. As a case in point, consider a dispute between two modal logicians defending, respectively, S4 and the Brouweresche system BR. The former maintains that S4 gets necessity right because it contains exactly the theses valid in all models consisting of possible worlds related by an accessibility relation which is reflexive and transitive. The latter begs to differ, saying that relative possibility is reflexive and symmetric. How is this to be resolved? Don't say: by looking at how the sentences match their ordinary language counterparts. We are considering here a dispute between model-theoretic necessitarians; that is, people who maintain that necessity is to be explained by the models. Needless to say, the attack on model-theoretic necessitarianism should not be construed as an attack on the discipline of model theory as such. If there is a nonnull class of necessary truths, then presumably there will be some class C

of models such that A is necessary iff A is true in all members of C; and it is the business of model theory to uncover such facts. My concern has been to cast doubt on the belief that the 'iff' *suffices* to furnish an explanation of the left hand side by the right hand side.

My broad point is recognised in a discipline close to modal logic, namely tense logic. The history of philosophy has seen many marvellous claims as to the necessary structure of time. Arthur Prior's pioneering work on the expressability of alternative classes of temporal structures by different classes of tense-logical theses, led naturally to the thought that none captured a necessary structure for time, so much as presented an alternative way time might be. The analogy isn't perfect; but it suggests not only that model theory does not by itself prove necessitarianism, but also that the existence of alternative models ought to make us less confident that any of them expresses necessary truth. The next section amplifies this last point.

4. UNDERSTANDING AND INTUITIONS

It is important to avoid the bewilderment response here: 'But I simply don't understand what 'true' and 'false' could amount to, if it is being proposed that there are models where one and the same linguistic item can be both, or various other logical laws are violated'. The short response is to urge the reader not to give way to such *semantic dogmatism*. There should of course be no quarrel with honest bewilderment, but that is far from dogmatic. The undogmatic response would be to consider seriously that such semantics amount to metaphysical proposals, of a highly general and theoretical kind, as to how the world might be.[13] The situation in 'logical physics' is simply no different from that in theoretical physics, only newer and more abstract. Uncriticisable dogmatism is as out of place in the former as the latter. Thus Popper: 'An argument that proceeds from inconceivability is, like other self evident arguments, always suspect'.[14]

It is all very well to make this point generally, but it seems to me that it helps noncomprehension to see some recent nonclassical logic. Model theory has a useful epistemic role to play, like that of mathematical physics; or differently, of fiction. It can feed into whatever faculty it is which enables us to conceive of false possibilities and contingencies, to expand those intuitions. So in this section I digress

into some recent technical work, and consider how it might aid incomprehension.

The *Law of Excluded Middle* and its relatives such as the Law of Bivalence, have come under a lot of pressure lately. Philosophers have often not found it so difficult to believe that some propositions might be neither true nor false. Brouwer and Heyting felt the same about mathematics, and Dummett has recently advanced a similar view about a broader class of propositions. My aim here is not to defend these views, so much as to invite the reader to ask himself or herself the question *what if they are right*? Good philosophy always attempts to understand one's opponent; and these writers have provided extensive descriptions of how they see the world, how the world *would* be if they *were* right. Needless to say, it is open to believe that the Law of Excluded Middle is *true*, without holding the extra thesis that it is *necessary*. And I suggest that in the presence of descriptions of how intuitionists imagine the world to be, it is difficult to feel confident about the *impossibility* of their view.

Recently, in the work of da Costa, Meyer, Priest, Routley, Rescher and Brandom and others, the truth of the *Law of Noncontradiction* has come under much fire. Like Excluded Middle, the Law of Noncontradiction is a group of theses, a representative being: no proposition is both true and false. Philosophers on the whole seem to find the breakdown of Noncontradiction harder to swallow than the breakdown of Excluded Middle, though it is well-known that the former in the form $\sim(A \,\&\, \sim A)$ is equivalent to the latter in the form $A \vee \sim A$, given only the independently plausible laws of Double Negation and de Morgan. A common case for giving up Noncontradiction is that it provides the most natural and unified treatment of the set-theoretic and semantic paradoxes, and logics exist which tolerate and contain the effect of contradictions by giving up the classical law *Ex Contradictione Quodlibet* $(A \,\&\, \sim A)/\therefore B$. In the course of the development of such logics it became clear that there is more than one position one could adopt. *Strong paraconsistentism* holds that some contradictions are true. *Weak paraconsistentism* holds that no contradictions are true, but that inconsistent 'worlds' need to be admitted to one's model. A version of weak paraconsistentism holds that inconsistent 'worlds' describe possibilities, so that while no contradictions are true, some contradictions are possible. The proof theory and model theory for

large classes of such logics is well developed and well discussed by now.

Now a common response by philosophers to the suggestion that a contradiction might be true, is *I don't understand*. The noncomprehension may be genuine, but it does not justify the often-accompanying rejection of the suggestion. One is reminded of philosophers' noncomprehension of the idea that space might be four-dimensional, or curved. There is, of course, a quick argument against noncomprehenders who also hold that contradictions are necessarily false: if you think they are false, then presumably you understand them. To be fair, the noncomprehension often unpacks as 'I don't know what it would be like for a contradiction to be true'. The remedy might well be to read the literature and to think about the way paraconsistentists imagine the world to be.

Sometimes, noncomprehension is accompanied by a manoeuvre which I call the *Three Monkeys' Decision*: the decision to use negation in such a way that contradictions cannot be true. I do not think that this decision is any more plausibly motivated than its conventionalist cousins. One such motivation is that one can determine meanings by decision, but the meanings of linguistic items are also linked to the role those items play in theory. So, it might be thought, there is a sense in which some propositions are impossible: the sense that if the accompanying theory is false then it needs terms with different meanings to describe that fact.[15] But this odd-sounding incommensurability thesis ought to alert us that something is wrong. For if the accompanying theory is false, then surely it is false as it is, with all its terms having whatever meanings their theoretical roles are capable of giving them. The trouble is, as it so often is, that a distinction has already been imported between the status of putatively necessary propositions and ordinary scientific propositions. Supposing that the meaning of ordinary scientific terms like 'electron' is tied to the role they play in theory, this does not ensure that the theories are true. They might be false, with no change in the meaning of 'electron'; and what then would we make of the 'decision' to use 'electron' such that the electronic theory of matter is true? This analogy has a further use. Let it be conceded that if a certain proposition turns out unexpectedly to be false, the new theory needed to describe the world might employ terms with sufficiently different theoretical roles to make it worth

saying that they have different meanings. Commensurability has not been sacrificed, however, since we still assert that the old theory is false. But also if the old theory were not too bad, if it got things fairly right in a limited domain even down to its theoretical concepts, then we *might* find sufficient analogies between corresponding theoretical roles in the old and new theories to be worth using the same term for both. This goes as much for negation as for electrons.

Perhaps the reader is comforted by the thought that there is at least one proposition which is absolutely irrefutable, the Law of Propositional Identity, $A \rightarrow A$. This brings me to the topic of *Martin's Theorem*. Martin's Theorem says that in a certain weak propositional calculus, no instance of $A \rightarrow A$ is provable.[16] The proof of the theorem proceeds by showing for given any instance of $A \rightarrow A$, how to construct a model in which it is false. The models are very abstract structures, as is not uncommon in algebraic and operational semantics, and do not on the face of it look much like the spacetime manifolds some have imagined possible worlds to be. And that is a significant feature of the present position: why couldn't what there is be an abstract structure? Philosophers have not found so much difficulty in supposing that abstract entities exist, though obviously I am not claiming that they *do* exist. Furthermore, some recent theorists have suggested that the existence of universals might be contingent,[17] and have proposed an account of contingent laws of nature on that basis. If you are one of these, then give serious consideration to the possibility that contingent, abstract entities might have been *all* that exist. If a way for $A \rightarrow A$ to be false is that the world be very different from the way it is now, then why not? Why *couldn't* the world have been nothing but (say) some three-valued semantical algebra? This point is reinforced by the observation that it is important not to confuse *generality* with *abstractness*. Our world has very general structural features too, for instance very general aspects of its differential topology. It is possible to present General Relativity, Quantum Mechanics, Gauge Theory, even Newtonian Dynamics in very abstract fashion. Considered in isolation from the concrete universe out of which they arise, it can be difficult to grasp their connection with our world. I suggest that things might well be that way with abstract-looking logical countermodels too. Something might be a universe not-too-dissimilar from our own, yet have struc-

tural aspects which render false all manner of propositions such as $\sim(A \,\&\, \sim A)$ or $A \rightarrow A$. There is, I suggest, no reason why such very general or abstract structures should not be realised. And if so, then they represent false-but-possible states of affairs.

The importance of Martin's Theorem is considerable, I think, since it bears on well-entrenched intuitions about analyticity. It is seductively easy to believe that 'If Smith is a bachelor, then Smith is an unmarried man' can be made true by the decision to use 'bachelor' interchangeably with 'unmarried man'. But it would take more than that to make it necessary. It needs 'If Smith is a bachelor then Smith is a bachelor' to be necessary also, and that is a substitution instance of $A \rightarrow A$. Indeed, it seems to me that the intuitive solidity of mathematics rests on the same foundation. Short, quite obvious inferences in mathematics often derive, like the previous bachelor case, from some definitional decision to use terms interchangeably applied to $A \rightarrow A$, (or to $(A \,\&\, B) \rightarrow A$ or $A \rightarrow (A \vee B)$).[18] Mathematical connections established by longer chains of reasonings appealing to more complex deductive principles are to that extent less evidently necessary. I am not suggesting here that it is *easy* to understand how standard mathematics might have been false. But then we should beware of projecting the limitations of our imaginations onto the world. The easiest understanding I am able to offer here is of the order of difficulty of whatever would make $A \rightarrow A$ false; and that, as has already been noted, looks to be pretty strange stuff.

I trust that these examples have been sufficient to shake the reader's confidence that any proposition is impossible. But note that they share the feature of arising at the level of propositional logic. So we should also look inside atomic propositions, to see if necessity might arise from sub-atomic structure. Here, as elsewhere in this paper, the weakness of the imagination limits consideration of every candidate for necessity which might be put up. But I do want to show some possibilities connected with the Law of Identity, everything is self-identical. I report here the work of various theorists, including Brady, da Costa, Meyer and others.

For instance, what would it be like for everything to be self-identical but also some things to be non-self-identical? That would be an inconsistent universe, but that should be the last of our worries by now. We can take set theory as our model for this possibility. Suppose

we take identity to be: every member of the one is a member of the other and vice versa; and non-identity to be: $x \neq y_{\mathrm{df}} (\exists x)$ $(z \in x \,\&\, \sim z \in y)$. Now consider an inconsistent set theory[19] containing the Law of Identity $(x)(x = x)$ and containing in addition the Russell set R. It takes only a few principles to show, as Russell did, that $R \in R$ & $\sim R \in R$. Whence $(\exists z)$ $(z \in R \,\&\, \sim z \in R)$, i.e. $R \neq R$. There are variants of this for different accounts of identity, which exploit variants of the Russell Paradox, e.g. (i) $x = y =_{\mathrm{df}} (\forall z)$ $(x \in z \equiv y \in z)$ and $x \neq y\, y =_{\mathrm{df}} (\exists z)$ $(x \in z \,\&\, \sim y \in z)$. Again, the Russell Set with $R \in R$ & $\sim R \in R$ gives $R \neq R$, (ii) $x = y =_{\mathrm{df}} (\forall F)$ $(Fx \equiv Fy)$ and $x \neq y_{\mathrm{df}} (\exists F)$ $(Fx$ & $\sim Fy)$. For F, take '$\in R$', (iii) $x = y =_{\mathrm{df}} (\forall u)$ (x participates in $u \equiv y$ participates in u). Then, supposing it were true (why not!) that $(\exists u)(\forall x)$ (x participates in $u \equiv Fx$), we have $u = u \,\&\, u \neq u$. A somewhat different approach develops inconsistent number theories,[20] according to which (x) $(x = x)$ but also $O \neq n$ for some n. This produces $O \neq O$ and, indeed, $n \neq n$ for every n so that (x) $(x \neq x)$ as well. These structures do not have every proposition true in them, since $O = 1$ is true in none of them. Indeed in some such structures, distinct numbers are distinct so that $n \neq m$ for distinct n, m. In fact it is possible to produce such structures in which the domain is divided into two disjoint classes, those for which $n = n$ but not also $n \neq n$, and those for which $n \neq n$ but not also $n = n$; which gives a universe in which some things are self-identical and some (other) things are non-self-identical. Again, a different approach might take self-identity as the criterion of existence. This is not uncommon; recall the definition of the null set $\Lambda =_{\mathrm{df}} \{x : x \neq x\}$. This gives the possibility of a universe in which nothing is self-identical and everything is non-self-identical. Indeed, this universe seems to be consistent. At least, it is consistent if it is consistent that nothing exists, which has seemed a desirable proposition to some.

The aim in this section has been to outline recent technical experimentation which, at the very least, has to be recognised as exploring the limits of conceivability. In the absence of successful general arguments for setting those limits narrowly, one has to take a tolerant attitude to which abstract and general theories count as possible, a practice which is entirely uniform with theoretical science. Needless to say, it is the aim of this paper to criticise attempts at such general arguments; and in the next section, a group of such arguments are considered.

5. A PRIORI ARGUMENTS FOR NECESSITARIANISM, AND MINIMAL METALINGUISTIC CONSISTENCY

In 'There Is At Least One A Priori Truth', Hilary Putnam tries out, then rejects, an argument to the effect that logical truth is needed to constitute rationality.[21] Since he is attracted by the epistemic link between fallibilism and possibilism, he proposes the view that every statement is revisable in the sense of 'challenging a concept it contains', but that it is not the case that the rational revision of every statement permits denying it. Evidently this is an attempt to save some logic. Putnam notes that it echoes Quine's well-known translatability-of-the-connectives argument, which I take to be a sophisticated version of the 'I don't understand' argument criticised earlier. Putnam is inclined to believe the argument that it would follow that it is a necessary truth that if the relevant concepts are not revised, then P, for suitable P. The correct strategy against Putnam here would seem to be to deny the premiss of the argument, that the rational revision of some statements does not permit denying them. It has already been argued that such denial can be *intelligible*, and there does not seem to be any obvious methodology to force us to admit that it is *irrational*.

Putnam is also worried by the thought that it looks pretty necessary to assert that *not every proposition is both true and false*. In a similar vein, it used to puzzle me whether the following might be a candidate for necessity: *at least one proposition fails to be true*. But I do not think so. What would the universe be like for that not to hold? That is easy to describe: every proposition would be true. Needless to say, that is not how things *are*. But necessarily so? We can *say* easily enough what it would be for every proposition to be true. It has to be admitted that it is not always easy to understand the claim that P is possible, for selected extreme P's, but it is good advice not to trust one's noncomprehensions, especially when there is comprehension here of a sort. Again, consider the proposition that *at least one proposition is true*. Of course that is true, but consider the following argument which might be made out for its necessity. Suppose it were false that at least one proposition is true. Then the proposition that it is false that at least one proposition is true, would be true. Hence at least one proposition is true. Hence, it is necessary. What I think can be said against this argument is that it turns on the principle $\sim A \rightarrow A / \therefore \Box A$. While this would be plausible if the $\rightarrow$ in question were entailment, it should not

tempt anyone if the $\rightarrow$ is some ordinary (nonentailmental) 'if . . . then'. It fails, for instance, if the $\rightarrow$ is $\supset$. One would need a further necessitarian premiss if one is not to be accused of simply pulling a necessitarian conclusion out of the hat.

One might wonder whether there is a transcendental argument for some sort of metalinguistic controls which amount to necessity. With any assertion we make, for instance the assertions in this paper, we presumably also do not wish to put forward its denial. This suggests a principle of *minimal metalinguistic consistency*[22] as a constraint on assertibility or perhaps rationality; or at the *very* least minimal metalinguistic *nontriviality*, that is that one could hardly intelligibly entertain all one's assertions and *all* their denials. Now I do not think that this is an easy argument to come to grips with, though it must be agreed that it touches on very deep intuitions about assertibility and intelligibility. The situation is complicated by the fact that, *pace* Tarski, it is notoriously difficult to make out a case for a rigid distinction between object language and metalanguage for useful natural languages, only for artificial languages which fall short of reality. So one might be inclined to conclude that asserting any contradiction violates a metalinguistic constraint on assertibility and so intelligibility. The further conclusion is thus that contradictions are unassertible, thus unintelligible, and so their denials are necessarily true.

I think that the argument here is better off aiming for the weaker conclusion that one's theory should be minimally *nontrivial*, rather than the stronger conclusion that it should be *consistent*. As to the latter, I would agree that in asserting A one does not also typically intend to deny it, but it might be that one's conceptual discoveries can *surprise* one. One might find that there are isolated instances where one is *forced* to contemplate inconsistency: consider recent paraconsistentist work on inconsistent solutions to the Liar Paradox and Russell's Paradox, or on the possibility that the empirical science of motion might need to be inconsistent. Consistency, even metalinguistic consistency, is one among many theoretical desiderata, and fallibilism should recognise that consistency might occasionally best be sacrificed in the interests of overall theoretical economy. That is to say, *our concepts escape our control*: we may seek to quarantine the true from the false, but discover that this is not entirely achieved in the best theory available. To repeat an earlier point, this does not

mean that 'true' and 'false' have lost all their meaning, since they may well retain strong theoretical links and parallels with the Tarskian concepts, sufficient for it to be worth viewing as a theoretical discovery about the concepts of truth and falsity. The same observations apply to minimal metalinguistic nontriviality. Remember that it is not being maintained that every proposition and its denial *are* true, only that it might have been so. It would be hard to make out a case for such a universe to be *interesting*, and just as hard to envisage anyone seriously asserting that it is our universe. On the other hand, it would seem to be that in *our* universe the conditions for the minimal intelligibility and assertibility of the proposition of the nontriviality of the world are fulfilled, and thus the conditions for the intelligibility of the denial of that proposition would also seem to be met. The latter does seem to be in accord with intuition.

6. CONCLUSION

We have seen that some, though not all, of the considerations urged against various versions of necessitarianism are epistemological in spirit. Now Susan Haack has pointed out that the doctrines of fallibilism and necessitarianism are formally consistent.[23] Whatever one makes of fallibilism, it is easy to assent to the proposition that good scientific method is consistent with the existence of at least one necessary truth. However, I maintain that one can sidestep these points by holding that a properly general epistemic monist conception of scientific method, fallibilist or not, gives no reason to think that any proposition is necessary and every reason to think not.

After all, consider what theoretical function fallibilist necessitarianism might serve. Why should one believe that at least one proposition is necessary? In the first place, it is hard to make sense of how our *perceptions* of the world might make for a difference between two different sorts of truths, sufficient to dignify one kind as necessary and the other not. We do not perceive, with any sense organ, anything more than that propositions are true, certainly not that they are necessary. There is no epistemic dualist split in perception.

Thus if necessity is to get an epistemic foot in the door at all, it should be conjectural or postulational. One would in one's theory conjecture of certain propositions not merely that they are true, but *also* that they are necessarily true. I suggest that the role of such

necessitarian elements in the theory would be, indeed is, to erect a barrier to legitimate criticism of the theory, to avoid having to back up one's position against critical scrutiny.

Ask yourself how often you encounter necessitarian elements in philosophical theory. I think that they are frustratingly ubiquitous. Philosophers are hooked on *a priori* certitudes like logical junkies, unable to discard the necessitarian needle. No doubt this is partly a hangover from the not-so-distant past when philosophy was thought of as discontinuous from science, as second order and essentially *a prioristic*. But such dualism ought to be seen to be as undesirable as it is unnecessary. I submit that ***necessitarian explanations are simply never needed in theorising about reality***. Putnam nibbles at this:

> Nor do we really need a proof that a statement is *a priori* in this sense (rationally revisable) very often. If a statement has the property that *we cannot now describe* any circumstances under which it would be rational to give it up, that will surely suffice for most purposes of philosophical argument.[24]

So I suggest that the present position would seem to have many interesting ramifications as epistemology is naturalised throughout philosophy, abolishing the hydra of necessitarianism. If a chain of responses to requests for reasons is stopped with the claim that a particular premiss is necessarily and self-evidently true, then nothing is added and no light is shed. '*P* because *Q*' may be helpful, but '*P* because necessarily *P*' is useless. '*P* because I can prove it' only invites the request to do so. Certainly human debate stops, but Putnam showed the correct stopping point: we stop when imagining an alternative is beyond us.

Summing up, nothing ensures that the principles that the formal logician chooses to follow are necessarily true, but then nothing prevents us from error in any case. A properly general and unified theory of the world is to be desired, and the present view seems to be the only candidate around.

NOTES

[1] In defence of these claims, see my paper with Tim Burgess [15]; see Popper [20] and Lakatos [8, 9]. On the term 'possibilism' see Naess' excellent [16] and Nerlich's [17].

[2] See Popper [20] p. 78, or [15]. Also in accord is Putnam's [22]. The point of the 'in principle' is to allow for the fact that not all conventions originate in actual decisions.

[3] See Wachbroit [28], pp. 48–9.

[4] Wachbroit [28], p. 50; Quine [23].
[5] Haack's [5] is a good account of these changing views.
[6] Quine [24]. Wachbroit presses Quine's objection in 'Truth by Convention' that no finite set of conventions can determine an infinite set of logical truths, and argues that the attempt to salvage this by allowing self-referring conventions falls foul of a diagonal argument. But if even one necessary truth is determined by a convention, necessitarianism is established; so that needs prevention, see [28], pp. 51–3.
[7] Popper: 'It is not the properties of the world which determines this construction [of science]; on the contrary it is this construction which determines the properties of an artificial world', [20], p. 79. For a discussion of indeterminateness in conventionalist semantics, see Nerlich [18] p. 100.
[8] Even so, it would be a distinctly odd thesis: in what sense would the world be indeterminate about whether all cats are cats?
[9] Putnam [22], p. 163.
[10] Even here, though there is a difficulty if 'not give up' amounts to 'continue to believe'; for then the idea that beliefs be decision-controlled amounts to the thesis that beliefs are actions, which is controversial.
[11] Cognate views are evidently the indeterminateness thesis discussed earlier, or some 'no proposition expressed' thesis.
[12] The remarks here are intended to apply as much to those necessitarians for whom proof precedes truth, as the other way around.
[13] See e.g. Routley, [26].
[14] [21], p. 207.
[15] Haack [5] argues against the thesis that there may be truths which are 'analytique' in the sense that everyone learns that they are true by learning the words, attributing it to the later Quine.
[16] Proved by Errol Martin in [10]; see also Martin and Meyer [11]. It constituted a positive answer to Belnap's Conjecture, that in the logic S, consisting of rule transitivity and axioms and rules of prefixing and suffixing, no instance of $A \rightarrow A$ is provable. Martin and Meyer have exploited the result to defend the view that Propositional Identity is not a logical law and that validity is captured exactly by the theory of the syllogism, but it seems to me that the latter conclusion is uwarranted.
[17] Armstrong [1], Tooley [27].
[18] It goes without saying that the latter two have been questioned: $A \,\&\, B \rightarrow A$ is denied in connexivism on which the literature is considerable, e.g. McCall [12]. Many find $A \rightarrow A \vee B$ of doubtful value, contrary to Haack; see Parry [19].
[19] Brady [2] or da Costa [3].
[20] [13] or [14].
[21] Putnam [22]. Popper runs a similar line, that logic is needed as an 'organon of criticism', and that classical logic is the preferred logic since it alone maximises retransmission of falsity. This argument is discussed and rejected in [15].
[22] I am indebted for raising these points to Dr. Rainer Trapp, and a referee of this journal.
[23] Haack [6] and [7].
[24] [22], p. 170.

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